

Rearranging Formulas

One universal method: clean the formula, then isolate the subject. It works when the subject appears once, twice, or under a root.



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THE TOPIC ON ONE MAP

Rearranging Formulas

01 Subject once

"make t the subject"

clean, then isolate

02 Powers and roots

$\sqrt{\quad}$ round a side: square

t^2 : $\pm\sqrt{\quad}$

03 Subject twice

collect $\rightarrow$ factorise $\rightarrow$ divide

04 Root and twice

$\sqrt{\frac{x+a}{x-b}}$

05 Quadratic in the subject

complete the square first

$f^{-1}(x)$

THE UNIVERSAL METHOD

PRELIMINARY

Root / power?
undo it

STAGE 1

Clean
brackets · fractions

Collect
subject one side

STAGE 2

Factorise
if it is twice

Divide
by the bracket

01 Subject Once: Clean, then Isolate

Undo what was done to the subject — in reverse order.

WHEN YOU SEE → YOU DO

“make t the subject”

→ clean, then collect and divide

“rearrange ... to make x the subject”

→ undo in reverse order

1

Clean

Multiply by any denominator. Expand a bracket if the subject is inside it.

2

Collect and divide

Subject term on one side, everything else on the other. Divide by what multiplies the subject.

WORKED MODEL • MAKE F THE SUBJECT

$$C = \frac{5(F - 32)}{9}$$

1 $9C = 5(F - 32)$

clean: $\times 9$

2 $9C = 5F - 160$

clean: expand

3 $5F = 9C + 160$

collect

ANSWER $F = \frac{9C + 160}{5}$

IDEA 01

Practise: Subject Once

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Collect, divide

CH 2 • TOPIC 7 2 marks

Make a the subject of $v = u + at$.

$a =$ _____

Q2 DR ESLAM Clean the fraction first

CH 2 • TOPIC 7 2 marks

Make x the subject of $y = \frac{x+4}{3}$.

$x =$ _____

SELECTED PRACTICE

Q3

PRACTICE

Practice • Easy

CH 2 • TOPIC 7

2 marks

- 1 Make h the subject of the formula

$$t = \frac{gh}{10}$$

(2 marks)

Q4

PRACTICE

Practice • Easy

CH 2 • TOPIC 7

2 marks

- 5 Make t the subject of the formula $p = at - d$

(2 marks)

Q5**PRACTICE**

Practice • Easy

CH 2 • TOPIC 7 2 marks

11 Rearrange $v = u + at$ to make t the subject of the formula.

(2 marks)

Q6**PRACTICE**

Practice • Medium

CH 2 • TOPIC 7 2 marks

12 Use the formula $s = ut + \frac{1}{2}at^2$.

Make a the subject of the formula.

 $a = \dots\dots\dots$

(2 marks)

PAST PAPERS

Q7**PAST PAPER**

Edexcel 4MA1 · January 2020 · Paper 1HR · Q3

CH 2 · TOPIC 7 8 marks

3 (a) Make a the subject of $d = g + 2ac$

.....

(b) Factorise fully $9ef - 12f$

.....

(c) Expand and simplify $(x + 2)(x - 5)$

.....

(d) Simplify fully $\frac{n^4 \times n^7}{n^5}$

.....

(Total for Question 3 is 8 marks)

Q8**PAST PAPER**

Edexcel 4MA1 · January 2021 · Paper 1HR · Q6

CH 2 · TOPIC 7 5 marks

6 (a) Make c the subject of $A = \frac{c}{y} - 5z$

.....

(b) Write down the value of g^0

.....

(c) Factorise $x^2 - 11x + 24$

.....

(Total for Question 6 is 5 marks)

Q9**PAST PAPER**

Edexcel 4MA1 · May 2021 · Paper 1H · Q5

CH 2 · TOPIC 7 6 marks

5 (a) Expand and simplify $3x(2x + 3) - x(3x + 5)$

.....

(b) Make t the subject of the formula $p = at - d$

.....

(2)

Given that $\frac{w^5 \times w^n}{w^3} = w^{10}$

(c) work out the value of n .

 $n =$ **(Total for Question 5 is 6 marks)**

02 Powers and Roots: the Preliminary Step

A root or a square around the subject comes off **first** — or **last**, whichever layer is outside.

WHEN YOU SEE → YOU DO

$\sqrt{\quad}$ around a whole side

→ square both sides first

subject squared, t^2

→ $\pm\sqrt{\quad}$ at the end

Whole side under $\sqrt{\quad}$ →

square both sides

Subject squared, t^2 →

$\pm\sqrt{\quad}$ both sides (at the end)

Cube root / cube →

cube / cube-root both sides

WORKED MODEL • ROOT OUTSIDE

$$m = \sqrt{\frac{k+1}{4}}$$

$$1 \quad m^2 = \frac{k+1}{4}$$

square both sides

$$2 \quad 4m^2 = k+1$$

× 4

ANSWER $k = 4m^2 - 1$

× **TRAP** Square-rooting gives $\pm$: from $y = 3p^2 - 4$, $p = \pm\sqrt{\frac{y+4}{3}}$.

IDEA 02

Practise: Powers and Roots

My questions first, then selected practice.

Q10 DR ESLAM Squared subject

CH 2 · TOPIC 7 2 marks

Make r the subject of $A = \pi r^2$.

Divide first, then square-root.

$r =$ _____

Q11 DR ESLAM Root in the middle

CH 2 · TOPIC 7 3 marks

Make h the subject of $T = 2\pi\sqrt{\frac{h}{g}}$.

$h =$ _____

SELECTED PRACTICE

Q12**PRACTICE**

Practice · Easy

CH 2 • TOPIC 7 2 marks

9 Make t the subject of $c = t^3 - 8v$

(2 marks)

Q13**PRACTICE**

Practice · Medium

CH 2 • TOPIC 7 3 marks

1 Make y the subject of the formula $p = \sqrt{\frac{x+y}{5}}$

(3 marks)

Q14 PRACTICE Practice · Medium

CH 2 · TOPIC 7 3 marks

4 Make p the subject of the formula $y = 3p^2 - 4$

(3 marks)

Q15 PRACTICE Practice · Medium

CH 2 · TOPIC 7 2 marks

9 Make t the subject of $s = \frac{1}{2} at^2$

(2 marks)

Q16**PRACTICE**

Practice · Medium

CH 2 · TOPIC 7 2 marks

13 Make t the subject of $s = \frac{1}{2} at^2$

(2 marks)

Q17**PRACTICE**

Practice · Hard

CH 2 · TOPIC 7 3 marks

13 Rearrange $y = \frac{1}{\sqrt{x+1}}$ to make x the subject.

(3 marks)

03 Subject Twice: Factorise

The key trick. When the subject is in two terms, collect them and take the subject out as a common factor.

WHEN YOU SEE → YOU DO

subject in two terms

→ collect → factorise → divide

1

Clean

× the denominator, then expand so the subject is free.

2

Collect

All subject terms on one side, everything else on the other.

3

Factorise and divide

Subject outside a bracket → divide by the bracket.

WORKED MODEL • NOV 2021 P1H

$$w = \frac{2f + 3}{8 - f}$$

1 $w(8 - f) = 2f + 3$

clean: × (8 - f)

2 $8w - wf = 2f + 3$

expand

3 $8w - 3 = 2f + wf$

collect f terms

4 $8w - 3 = f(2 + w)$

factorise

ANSWER $f = \frac{8w - 3}{w + 2}$

IDEA 03

Practise: Subject Twice: Factorise

My questions first, then selected practice, then real past papers.

Q18 DR ESLAM Twice, no fractions

CH 2 • TOPIC 7 2 marks

Make x the subject of $ax + b = cx + d$.

$x =$ _____

Q19 DR ESLAM Twice, with a fraction

CH 2 • TOPIC 7 3 marks

Make x the subject of $y = \frac{x+2}{x-3}$.

× $(x - 3)$ · expand · collect x · factorise · divide.

$x =$ _____

SELECTED PRACTICE

Q20

PRACTICE

Practice • Medium

CH 2 • TOPIC 7 3 marks

2 Make t the subject of the formula $2(d - t) = 4t + 7$

(3 marks)

Q21

PRACTICE

Practice • Hard

CH 2 • TOPIC 7 3 marks

2 Make v the subject of the formula $w = \frac{15(t - 2v)}{v}$

(3 marks)

Q22**PRACTICE**

Practice · Hard

CH 2 · TOPIC 7 4 marks

5 Make a the subject of the formula $p = \frac{3a+5}{4-a}$

(4 marks)

Q23**PRACTICE**

Practice · Hard

CH 2 · TOPIC 7 3 marks

8 Make m the subject of the formula $f = \frac{3m+4}{m-1}$

(3 marks)

PAST PAPERS

Q24 PAST PAPER Edexcel 4MA1 · November 2021 · Paper 1H · Q17

CH 2 · TOPIC 7 8 marks

17 (a) Factorise $6y^2 - y - 5$

(2)

(b) Make f the subject of $w = \frac{2f + 3}{8 - f}$ (c) Express $4x^2 - 8x + 7$ in the form $a(x + b)^2 + c$ where a , b and c are integers.

(Total for Question 17 is 8 marks)

Q25**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 2H • Q14

CH 2 • TOPIC 7 7 marks

- 14 (a) Expand and simplify $(5 - x)(2x + 3)(x + 4)$
Show your working clearly.

(b) Make c the subject of $g = \frac{c + 3}{4 + c} - 7$

(Total for Question 14 is 7 marks)

Q26**PAST PAPER**

Edexcel 4MA1 • November 2023 • Paper 1H • Q15

CH 2 • TOPIC 7 3 marks

- 15 Make n the subject of the formula $x = \frac{3p + n}{3n - 4}$

(Total for Question 15 is 3 marks)

04 The Full Exam Question: Root and Twice

10 of the 16 past papers here are this one type. It is Idea 02 and Idea 03 joined together.

WHEN YOU SEE → YOU DO

$$\sqrt{\frac{\dots X \dots}{\dots X \dots}}$$

→ square → × denominator → collect → factorise

WORKED MODEL • JAN 2020 P2H

$$y = \sqrt{\frac{x+1}{x-4}}$$

$$1 \quad y^2 = \frac{x+1}{x-4}$$

preliminary: square

$$2 \quad y^2(x-4) = x+1$$

clean: × denominator

$$3 \quad xy^2 - 4y^2 = x+1$$

expand

$$4 \quad xy^2 - x = 4y^2 + 1$$

collect x terms

$$5 \quad x(y^2 - 1) = 4y^2 + 1$$

factorise

ANSWER $x = \frac{4y^2 + 1}{y^2 - 1}$

CHECK Pick $x = 5$: $y = \sqrt{6}$, so $y^2 = 6 \rightarrow \frac{4(6)+1}{6-1} = 5$. Correct.

IDEA 04

Practise: Root and Twice

My questions first, then selected practice, then real past papers.

Q27**DR ESAM**

The full exam type

CH 2 • TOPIC 7 4 marks

Make x the subject of $p = \sqrt{\frac{x+3}{x-1}}$.

$x =$ _____

SELECTED PRACTICE**Q28****PRACTICE**

Practice • Hard

CH 2 • TOPIC 7 4 marks

10 Make c the subject of the formula $p = \sqrt{\frac{ac+8}{3+c}}$

(4 marks)

Q29**PRACTICE**

Practice · Hard

CH 2 · TOPIC 7 4 marks

12 Given that $n > 0$ make n the subject of the formula $y = \frac{n^2 + d}{n^2}$

(4 marks)

Q30**PRACTICE**

Practice · Hard

CH 2 · TOPIC 7 4 marks

14 Make e the subject of $w = \sqrt{\frac{e+g}{ef-d}}$

(4 marks)

PAST PAPERS

Q31**PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 2H • Q16

CH 2 • TOPIC 7 4 marks

16 Make x the subject of $y = \sqrt{\frac{x+1}{x-4}}$

(Total for Question 16 is 4 marks)

Q32**PAST PAPER**

Edexcel 4MA1 · January 2021 · Paper 2H · Q14

CH 2 · TOPIC 7 7 marks

14 (a) Solve $\frac{9a-7}{5} - \frac{3a-7}{4} = 4.55$

Show clear algebraic working.

$a = \dots\dots\dots$

(b) Make c the subject of the formula $p = \sqrt{\frac{ac+8}{3+c}}$

$\dots\dots\dots$

(Total for Question 14 is 7 marks)

Q33**PAST PAPER**

Edexcel 4MA1 · January 2022 · Paper 2HR · Q15

CH 2 · TOPIC 7 4 marks

15 Make t the subject of $n^2 = \frac{4d+t^3}{t^3}$

$\dots\dots\dots$

(Total for Question 15 is 4 marks)

Q34 **PAST PAPER** Edexcel 4MA1 • June 2023 • Paper 2HR • Q17**CH 2 • TOPIC 7** **4 marks**

17 Make x the subject of $y = \sqrt[3]{\frac{6+5x}{x+4}}$

(Total for Question 17 is 4 marks)

Q35 **PAST PAPER** Edexcel 4MA1 • June 2023 • Paper 2H • Q17**CH 2 • TOPIC 7** **7 marks**

17 (a) Expand and simplify $(x+6)(3x-2)(x+6)$

(b) Make e the subject of $w = \sqrt{\frac{e+g}{ef-d}}$

(Total for Question 17 is 7 marks)

Q36**PAST PAPER**

Edexcel 4MA1 • June 2024 • Paper 2H • Q15

CH 2 • TOPIC 7 7 marks

15 (a) Make g the subject of $e = \sqrt{\frac{7g+5}{11+2g}}$

(b) Solve the inequality $3y^2 + 4y - 32 > 0$
Show your working clearly.

(Total for Question 15 is 7 marks)

Q37**PAST PAPER**

Edexcel 4MA1 · November 2024 · Paper 2H · Q12

CH 2 · TOPIC 7 7 marks

- 12 (a) Solve $\frac{3x+2}{5} - \frac{2x+1}{3} = x$
Show clear algebraic working.

$$x = \dots\dots\dots (3)$$

- (b) Make c the subject of the formula $f = \sqrt{\frac{a+bc}{c-d}}$

(Total for Question 12 is 7 marks)**Q38****PAST PAPER**

Edexcel 4MA1 · June 2025 · Paper 2HR · Q17

CH 2 · TOPIC 7 4 marks

- 17 Make k the subject of $p = \frac{8k^2+5}{7-3k^2}$

(Total for Question 17 is 4 marks)

Q39**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 2H • Q16

CH 2 • TOPIC 7 4 marks

16 Make t the subject of the formula $c = \frac{t^2 + 3}{7 - 8t^2}$

(Total for Question 16 is 4 marks)

Q40**PAST PAPER**

Edexcel 4MA1 • November 2025 • Paper 1H • Q18

CH 2 • TOPIC 7 4 marks

18 Make y the subject of $p = \sqrt{\frac{y + w}{3y - t}}$

(Total for Question 18 is 4 marks)

05 Quadratic in the Subject: Complete the Square

The subject appears as x^2 **and** as x ? You cannot collect them. The first step is **completing the square** — then x appears only once.

WHEN YOU SEE → YOU DO

subject as x^2 and x

→ complete the square first

“find $f^{-1}(x)$ ” of a quadratic

→ complete the square, then $\pm\sqrt{\quad}$

1

Complete the square

List $a, b, c, \frac{b}{2a} \rightarrow a\left(x + \frac{b}{2a}\right)^2 - a\left(\frac{b}{2a}\right)^2 + c$. (Chapter 2 • Topic 5.)

2

Now it is “subject once”

Move the number, divide by a , square-root with $\pm$, then move the last number.

WORKED MODEL • MAKE x THE SUBJECT

$$y = x^2 + 6x + 2$$

$$1 \quad y = (x + 3)^2 - 9 + 2 = (x + 3)^2 - 7$$

complete the square

$$2 \quad y + 7 = (x + 3)^2$$

move the number

$$3 \quad \pm\sqrt{y + 7} = x + 3$$

root with $\pm$

ANSWER $x = -3 \pm \sqrt{y + 7}$

× **TRAP** Trying to “factorise out x ” from $x^2 + 6x$ leaves $x(x + 6)$ — the subject is still there twice. Complete the square instead.

IDEA 05

Practise: Quadratic in the Subject

Questions written to practise exactly this idea.

Q41 DR ESLAM Complete the square first

CH 2 • TOPIC 7 3 marks

Make x the subject of $y = x^2 - 4x + 1$.

Step 1: $y = (x - 2)^2 - ?$ Step 2: move the number, then $\pm\sqrt{}$.

$x =$ _____

Q42 DR ESLAM With $a \neq 1$

CH 2 • TOPIC 7 4 marks

Make x the subject of $y = 2x^2 + 8x - 1$.

$x =$ _____

Q43 **DR ESLAM** Inverse of a quadratic

CH 2 • TOPIC 7 3 marks

$f(x) = x^2 + 10x$ for $x \geq -5$. Find $f^{-1}(x)$.

$f^{-1}(x) =$ _____

06 Where You Will Use It

Rearranging is the same skill as solving an equation — and it comes back in three later topics.

WHEN YOU SEE → YOU DO

“find $f^{-1}(x)$ ”

→ $y = f(x)$, make x the subject, swap

“solve” a linear equation

→ the same clean → isolate method

Linear equations

Same method, numbers instead of letters.

Inverse functions

$y = f(x) \rightarrow$ make x the subject $\rightarrow$ swap.

Rearranging

Simultaneous equations

Make x or y the subject, then substitute.

Formulas in science

$v = u + at$, $s = ut + \frac{1}{2}at^2$

WORKED MODEL • INVERSE FUNCTION

$$f(x) = \frac{x+2}{x-3} \quad \text{Find } f^{-1}(x).$$

$$1 \quad y(x-3) = x+2 \Rightarrow xy - x = 3y + 2$$

clean • collect

$$2 \quad x(y-1) = 3y+2 \Rightarrow x = \frac{3y+2}{y-1}$$

factorise • divide

ANSWER $f^{-1}(x) = \frac{3x+2}{x-1}$

IDEA 06

Practise: Where You Will Use It

Questions written to practise exactly this idea.

Q44 DR ESLAM Inverse function

CH 2 • TOPIC 7 3 marks

$$f(x) = \frac{2x + 1}{x - 4}. \text{ Find } f^{-1}(x).$$

Write $y = f(x)$, make x the subject, then swap x and y .

$$f^{-1}(x) = \underline{\hspace{5cm}}$$

Q45 DR ESLAM Same method, numbers

CH 2 • TOPIC 7 2 marks

Solve $5(x + 3) = 3x - 4$.

$$x = \underline{\hspace{5cm}}$$

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|--|--|
| 00 | Preliminary | Root outside $\rightarrow$ square both sides. Subject squared $\rightarrow \pm\sqrt{\quad}$ at the end. |
| 01 | Stage 1: Clean | $\times$ every denominator. Expand the bracket holding the subject. |
| 02 | Stage 2: Isolate | Collect $\rightarrow$ Factorise (if twice) $\rightarrow$ Divide. |
| 03 | Exam type | Square $\rightarrow \times$ denominator $\rightarrow$ expand $\rightarrow$ collect $\rightarrow$ factorise $\rightarrow$ divide. |
| 04 | x^2 and x | Complete the square first — then x is there once. Root with $\pm$. |
| 05 | Check | Put a number in for the subject, work out the other letter, and go back. |

BOTTOM LINE Undo the root $\rightarrow$ clean $\rightarrow$ collect $\rightarrow$ factorise $\rightarrow$ divide.

Answers

Rearranging Formulas • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • SUBJECT ONCE

- | | | |
|-----------|--|---|
| Q1 | Dr Eslam • Collect, divide | $a = \frac{v - u}{t}$ |
| Q2 | Dr Eslam • Clean the fraction first | $x = 3y - 4$ |
| Q3 | Practice • Easy | $h = \frac{10t}{g}$ |
| Q4 | Practice • Easy | $t = \frac{p + d}{a}$ |
| Q5 | Practice • Easy | $t = \frac{v - u}{a}$ |
| Q6 | Practice • Medium | $a = \frac{2(s - ut)}{t^2}$ |
| Q7 | Edexcel 4MA1 • January 2020 • Paper 1HR • Q3 | (a) $a = \frac{d - g}{2c}$ (b) $3f(3e - 4)$ (c) $x^2 - 3x - 10$ (d) n^6 |
| Q8 | Edexcel 4MA1 • January 2021 • Paper 1HR • Q6 | (a) $c = y(A + 5z)$ (b) 1 (c) $(x - 3)(x - 8)$ |
| Q9 | Edexcel 4MA1 • May 2021 • Paper 1H • Q5 | (a) $3x^2 + 4x$ (b) $t = \frac{p + d}{a}$ (c) $n = 8$ |

IDEA 02 • POWERS AND ROOTS

Q10 Dr Eslam • Squared subject

$$r = \sqrt{\frac{A}{\pi}}$$

Q11 Dr Eslam • Root in the middle

$$h = \frac{gT^2}{4\pi^2}$$

Q12 Practice • Easy

$$t = \sqrt[3]{c + 8v}$$

Q13 Practice • Medium

$$y = 5p^2 - x$$

Q14 Practice • Medium

$$p = \pm \sqrt{\frac{y + 4}{3}}$$

Q15 Practice • Medium

$$t = \sqrt{\frac{2s}{a}}$$

Q16 Practice • Medium

$$t = \sqrt{\frac{2s}{a}}$$

Q17 Practice • Hard

$$x = \frac{1 - y^2}{y^2}$$

IDEA 03 • SUBJECT TWICE: FACTORISE

- Q18** Dr Eslam • Twice, no fractions $x = \frac{d - b}{a - c}$
- Q19** Dr Eslam • Twice, with a fraction $x = \frac{3y + 2}{y - 1}$
- Q20** Practice • Medium $t = \frac{2d - 7}{6}$
- Q21** Practice • Hard $v = \frac{15t}{w + 30}$
- Q22** Practice • Hard $a = \frac{4p - 5}{p + 3}$
- Q23** Practice • Hard $m = \frac{f + 4}{f - 3}$
- Q24** Edexcel 4MA1 • November 2021 • Paper 1H • Q17
 (a) $(6y + 5)(y - 1)$ (b) $f = \frac{8w - 3}{w + 2}$ (c) $4(x - 1)^2 + 3$
- Q25** Edexcel 4MA1 • June 2022 • Paper 2H • Q14
 (a) $-2x^3 - x^2 + 43x + 60$ (b) $c = -\frac{4g + 25}{g + 6}$
- Q26** Edexcel 4MA1 • November 2023 • Paper 1H • Q15
 $n = \frac{3p + 4x}{3x - 1}$

IDEA 04 • ROOT AND TWICE

Q27 Dr Eslam • The full exam type

$$x = \frac{p^2 + 3}{p^2 - 1}$$

Q28 Practice • Hard

$$c = \frac{8 - 3p^2}{p^2 - a}$$

Q29 Practice • Hard

$$n = \sqrt{\frac{d}{y - 1}}$$

Q30 Practice • Hard

$$e = \frac{dw^2 + g}{fw^2 - 1}$$

Q31 Edexcel 4MA1 • January 2020 • Paper 2H • Q16

$$x = \frac{4y^2 + 1}{y^2 - 1}$$

Q32 Edexcel 4MA1 • January 2021 • Paper 2H • Q14

$$(a) a = 4 \quad (b) c = \frac{8 - 3p^2}{p^2 - a}$$

Q33 Edexcel 4MA1 • January 2022 • Paper 2HR • Q15

$$t = \sqrt[3]{\frac{4d}{n^2 - 1}}$$

Q34 Edexcel 4MA1 • June 2023 • Paper 2HR • Q17

$$x = \frac{6 - 4y^3}{y^3 - 5}$$

Q35 Edexcel 4MA1 • June 2023 • Paper 2H • Q17

$$(a) 3x^3 + 34x^2 + 84x - 72 \quad (b) e = \frac{dw^2 + g}{fw^2 - 1}$$

Q36 Edexcel 4MA1 • June 2024 • Paper 2H • Q15

$$(a) g = \frac{5 - 11e^2}{2e^2 - 7} \quad (b) y < -4 \text{ or } y > \frac{8}{3}$$

Q37 Edexcel 4MA1 • November 2024 • Paper 2H • Q12

$$(a) x = \frac{1}{16} \quad (b) c = \frac{a + df^2}{f^2 - b}$$

Q38 Edexcel 4MA1 • June 2025 • Paper 2HR • Q17

$$k = \sqrt{\frac{7p - 5}{3p + 8}}$$

Q39 Edexcel 4MA1 • June 2025 • Paper 2H • Q16

$$t = \sqrt{\frac{7c - 3}{8c + 1}}$$

Q40 Edexcel 4MA1 • November 2025
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• Paper 1H • Q18

$$y = \frac{p^2t + w}{3p^2 - 1}$$

IDEA 05 • QUADRATIC IN THE SUBJECT

Q41 Dr Eslam • Complete the square first

$$y = (x - 2)^2 - 3 \Rightarrow x = 2 \pm \sqrt{y + 3}$$

Q42 Dr Eslam • With $a \neq 1$

$$y = 2(x + 2)^2 - 9 \Rightarrow x = -2 \pm \sqrt{\frac{y + 9}{2}}$$

Q43 Dr Eslam • Inverse of a quadratic

$$f(x) = (x + 5)^2 - 25 \Rightarrow f^{-1}(x) = -5 + \sqrt{x + 25}$$

IDEA 06 • WHERE YOU WILL USE IT

Q44 Dr Eslam • Inverse function

$$\frac{4x + 1}{x - 2}$$

Q45 Dr Eslam • Same method, numbers

$$x = -9.5$$